

DCIT 208 - Software Engineering

D7: Sprint 3 Review Report, **Security/Privacy/AI Audit and Observability** **Evidence**

Team: InnovateX

Project: ScentHub

Team Repository:

<https://github.com/selasi07/ScentHub->

Commit Hash:

a87edb9db8deecc32f4dc632a1866e5b48560196

Submission Date: 7 August 2026

A. Sprint Summary

A1. Sprint Goal and Dates Covered

Sprint Goal

The goal of Sprint 3 was to improve the reliability, security readiness, maintainability, and operational visibility of the ScentHub application before the release candidate stage.

The sprint focused on:

- Improving backend logging and failure visibility.
- Reviewing security practices within the application.
- Performing dependency vulnerability analysis.
- Improving repository security practices.

Sprint Period

Sprint: Sprint 3

Dates Covered: 1 August 2026– 7 August 2026

A2. Updated Sprint Backlog

Item	Status	Description
Improve backend logging	Completed	Added startup, route and error logging
Review database security	Completed	Checked database connection handling and SQL queries
Dependency security audit	Completed	Ran npm audit and resolved vulnerabilities
Repository security improvement	Completed	Added .gitignore and removed node_modules tracking
AI usage documentation	Pending	Update AI_USAGE.md
Staging deployment evidence	Pending	Provide deployment screenshot/link

A3. Implemented Functionality at End of Sprint

At the end of Sprint 3, ScentHub includes:

Backend Improvements

- Improved server startup logging.
- Improved database connection error reporting.
- Improved CRUD route visibility through logging.

Database Improvements

The system continues to use SQLite for inventory management.

Stored product attributes:

Attribute	Purpose
id	Unique product identifier
perfume_name	Product name
brand	Perfume manufacturer
category	Product classification
price	Product price
quantity	Available stock

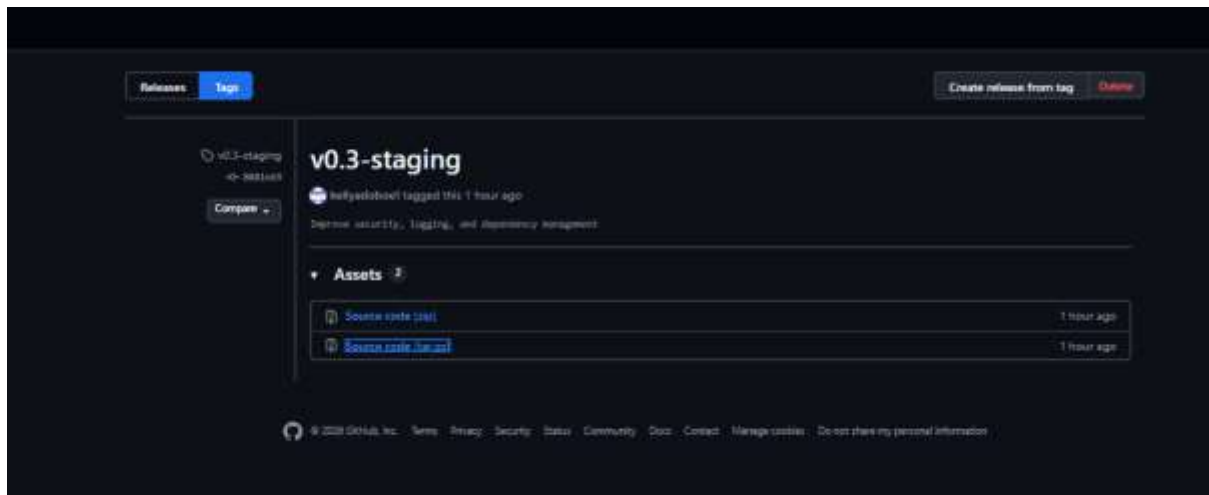
Security Improvements

Completed:

- Dependency vulnerability scanning.
- Dependency updates.
- Repository ignore rules.
- Security review of database operations.

A4. GitHub Evidence

Release Tag: v0.3-staging



B. Requirements and Backlog Changes

B1. Stories Changed During Sprint

During Sprint 3, no major functional user stories were added or removed.

The focus shifted from adding new features to improving system quality.

Changes made:

Change	Reason
Added logging improvements	Improve debugging and monitoring
Added dependency audit task	Identify security risks
Added repository security cleanup	Prevent unnecessary files being committed

B2. Traceability Impact

Existing product backlog functionality remained unchanged.

The implemented improvements support existing system requirements by improving reliability and maintainability.

C. Engineering Evidence

C1. Pull Requests and Review Evidence

Sprint 3 changes were implemented directly on the main development branch due to the project workflow used.

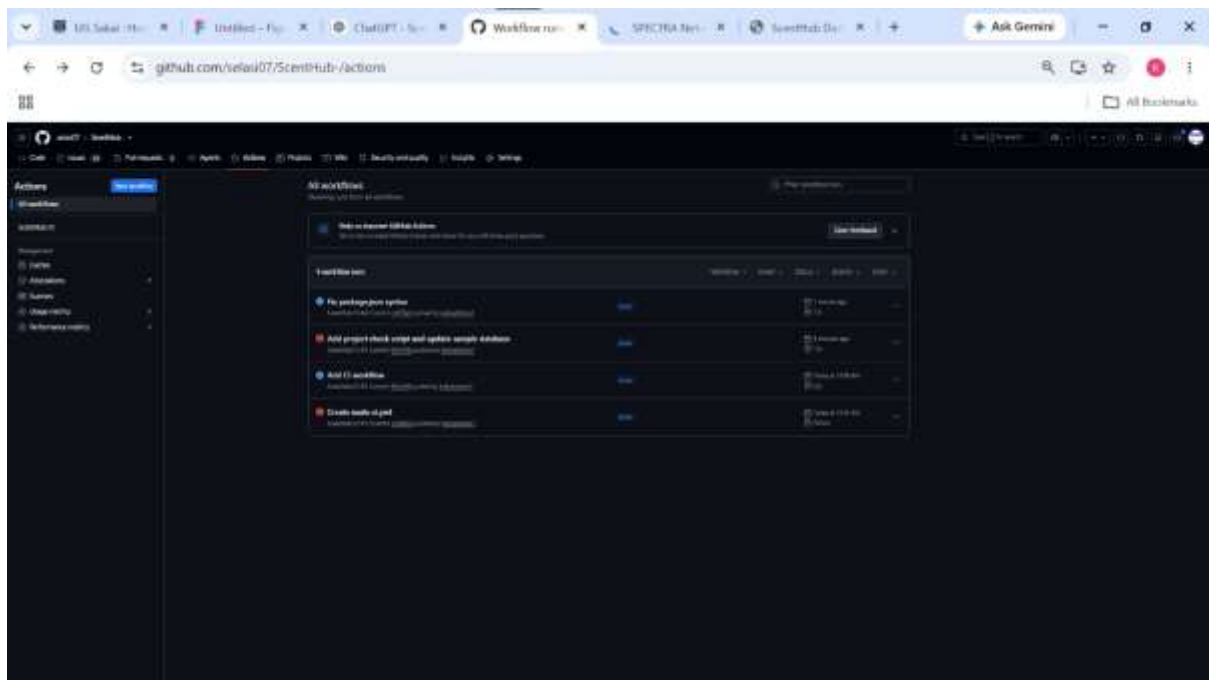
C2. Tests Added or Updated

Current testing evidence:

- Manual API testing performed.
- CRUD operations verified.
- Dependency audit performed.

C3. CI Evidence

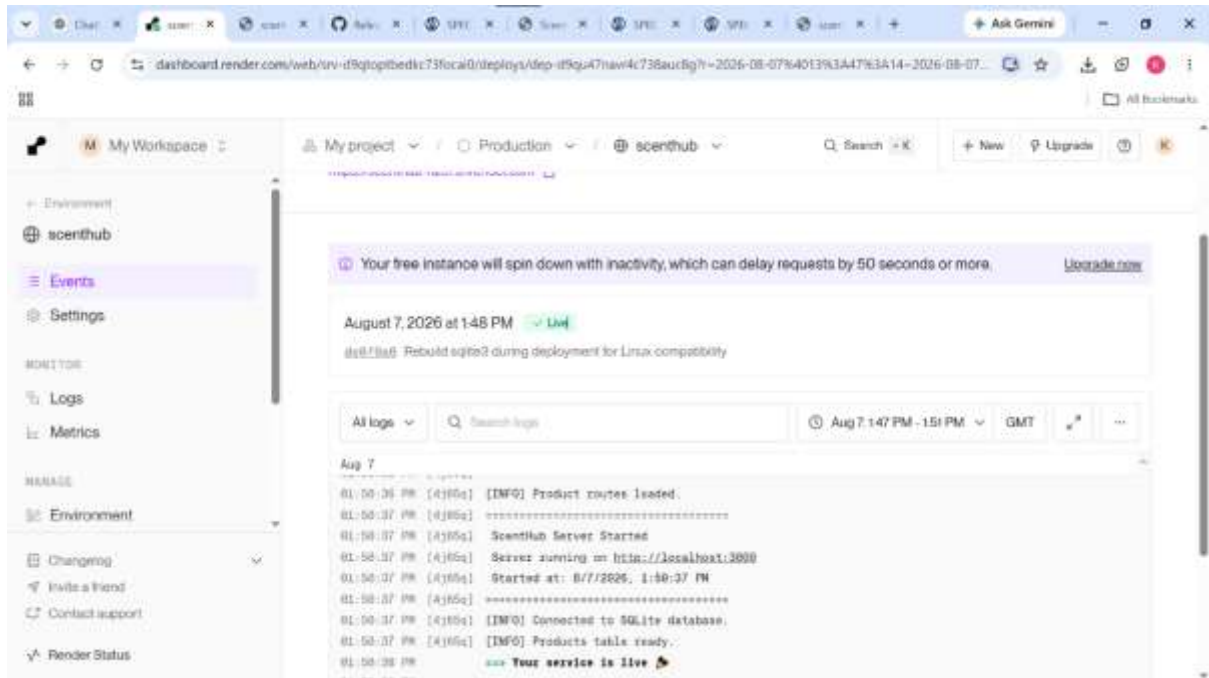
GitHub Actions was reviewed.



C4. Deployment Evidence

URL: <https://scenthub-h2er.onrender.com/>

Screenshot:



C5. Known Defects and Technical Debt

Known limitations:

Issue	Impact	Future Improvement
No authentication system	No user access control	Add login and roles
SQLite local database	Limited scalability	Move to production database
Limited automated tests	Less regression coverage	Expand test suite

D. Metrics and Reflection

D1. Story Points Planned vs Completed

Sprint 3 focused primarily on engineering quality improvements, security validation, and release readiness activities rather than introducing large new user-facing features.

Because the sprint concentrated on technical improvements, the team tracked work items as engineering tasks instead of using the same feature story-point estimation approach used in previous development sprints.

Sprint 3 Engineering Task	Status
Improve backend logging	Completed
Improve database error handling	Completed
Perform dependency security audit	Completed
Resolve dependency vulnerabilities	Completed
Add repository security improvements	Completed

Sprint 3 work was substantially completed, with remaining tasks focused mainly on final documentation and release evidence preparation.

D2. Team Velocity Comparison

Previous sprint velocities:

Sprint	Completed Story Points
Sprint 1	12
Sprint 2	22

Sprint 3 was a quality and stabilization sprint, therefore direct comparison using story points is not appropriate.

The team maintained development momentum by focusing on improving system quality, reliability, and release readiness.

D3. Sprint Reflection

Sprint 3 demonstrated a shift from feature implementation toward engineering maturity.

The team learned that:

- Security checks should be integrated throughout development.
- Logging is essential for identifying failures quickly.
- Dependency management should be performed regularly.
- Repository cleanliness improves maintainability.

D4. Lessons Learned

Sprint 3 provided several important lessons:

1. Security Reviews Should Be Continuous

The dependency audit revealed that security checks should not only be performed before release but should become part of the normal development process.

The team learned to regularly check:

- Third-party dependencies.
- Repository contents.
- Potential secret exposure.

2. Better Logging Improves Debugging

Before Sprint 3, diagnosing backend problems required manually checking application behaviour.

After implementing improved logging, developers can now identify:

- Server startup status.
- Database connection problems.
- API request activity.

This reduces debugging time and improves maintainability.

3. Repository Hygiene Matters

The discovery that node_modules was being tracked highlighted the importance of proper Git practices.

The team improved repository management by:

- Adding .gitignore.
- Removing generated dependency files.
- Keeping only necessary source files under version control.

D5. Changes Planned for Next Sprint

Future sprint improvements include:

- Completing staging deployment.
- Expanding automated testing.
- Adding authentication and authorization.
- Improving production monitoring.
- Continuing security reviews.

D6. Contribution Summary

Team Member	Contribution
Kelly Kofi Adoboe	Database/security improvements, dependency audit, logging improvements, documentation preparation

E. Client Feedback

E1. Client/User Review Evidence

No formal client feedback meeting was conducted during Sprint 3 since it was mainly about improving system quality and reliability. Internal review was used to evaluate improvements.

F. Security and Privacy Audit

F1. Threat Model

Threat	Risk	Mitigation
SQL Injection	Malicious database commands	Parameterized SQL queries
Dependency vulnerabilities	Exploitable packages	npm audit and updates
Secret exposure	Credentials leaked	.gitignore and repository review
Unauthorized access	Data modification	Future authentication system
Data loss	Inventory information lost	Database backup strategy

F2. Authentication and Authorization Plan

Current status:

ScentHub does not currently implement authentication.

Future plan:

- User login system.
- Role-based permissions.
- Restricted administrative actions.

F3. Input Validation

Implemented controls:

- Required product fields checked.
- Price validation.
- Quantity validation.
- Parameterized database queries.

F4. Secrets Management

Security review confirmed:

- No passwords stored.
- No API keys stored.
- No tokens stored.

Repository protection:

1. node_modules/
2. .env
3. database/*.db
4. *.log

F5. Dependency Risk Management

Process:

1. Run npm audit.
2. Review vulnerabilities.
3. Apply updates.
4. Verify again.

Evidence

Initial:

2 vulnerabilities

Final:

found 0 vulnerabilities

F6. Data Privacy Table

Data	Purpose	Sensitivity	Storage	Access
Perfume name	Inventory tracking	Low	SQLite database	System users
Brand	Product identification	Low	SQLite database	System users
Category	Classification	Low	SQLite database	System users
Price	Inventory value	Medium	SQLite database	Administrators
Quantity	Stock tracking	Medium	SQLite database	Administrators

G. Responsible AI Audit

G1. AI Usage Summary

AI tools were used for:

- Project planning.
- Code review suggestions.
- Documentation suggestions.
- Security review brainstorming.

G2. Verification Process

All AI-assisted suggestions were:

- Reviewed manually.
- Adjusted to fit the ScentHub Requirements.
- Verified before use.

G3. Confidential Data Statement

No confidential client information, passwords, API keys, or private credentials were provided to public AI tools.

G4. AI Risks and Mitigation

Risk	Mitigation
Incorrect suggestions	Manual review and testing
Security mistakes	Human verification
Over-reliance on AI	Developer approval required

H. Observability Evidence

H1. Logging Plan

Logged events:

- Server startup.
- Database connection.
- API requests.
- Application errors.

Not logged:

- Passwords.
- Secrets.
- Sensitive user information.

H2. Monitoring Evidence

Local monitoring evidence:

Example:

[INFO] Connected to SQLite database.

[INFO] Products table ready.

[INFO] Server running on http://localhost:3000

Insert screenshot.

H3. Failure Detection

The team can identify failures through:

- Server startup errors.
- Database connection errors.
- API request logs.

H4. Reliability Limitations

Current limitations:

- No cloud monitoring.
- No automated alert system.
- Local SQLite deployment.

Future improvements:

- Production monitoring.
- Cloud logging.
- Automated alerts.